

Digital twins for non-invasive tumor localization in breast-conserving surgery: a technical pilot study

Rafaela Timóteo^{a,b}, Alexandre Laborde^e, Bruno Vaz^e, Nuno Loução^b,
Yasna Forghani^{a,b}, André Cardoso^b, Mariana Correia^b, João Santinha^{a,b},
Daniel Simões Lopes^{c,d}, Tiago Marques^{a,b,1,*}, Pedro Gouveia^{a,b,1,**}

^a School of Medicine, University of Lisbon, Av. Prof. Egas Moniz MB, 1649-028, Lisboa, Portugal

^b Breast Cancer Research Program, Champalimaud Foundation, Av Brasília, 1400-038, Lisboa, Portugal

^c ITI/LARSyS, Av. Prof. Gama Pinto 2, 1649-003, Lisboa, Portugal

^d Instituto Superior Técnico, Universidade de Lisboa, Av. Rovisco Pais, 1049-001, Lisboa, Portugal

^e IT People, Praça Alvalade 7, 1700-345, Lisboa, Portugal

ARTICLE INFO

Keywords:

Digital twins
Extended reality
Breast-conserving surgery
Tumor localization

ABSTRACT

Inaccurate tumor localization in breast-conserving surgery (BCS) can lead to positive margins and re-excision. In this technical pilot study, Digital Twins (DT) were used for non-invasive tumor localization in the operating room. Breast DTs were generated from supine breast magnetic resonance imaging and registered to patients in real-time using a 3D sensing system during four surgeries. Alignment accuracy was measured via nipple deviations (0–5 mm). Setup and marking times were clinically acceptable, with high perceived usability (SUS = 95). These preliminary results demonstrate the potential of integrating DTs into surgical workflows.

1. Introduction

Breast cancer (BC) is the second most common cancer worldwide, with 2.3 million new cases reported annually [1,2]. Breast-conserving surgery (BCS) followed by adjuvant radiotherapy is the standard of care for early BC [3]. Early detection often involves small, non-palpable lesions, and accurate tumor localization is critical to achieve clear margins and minimize re-excision [4,5]. Surgical planning typically relies on prone magnetic resonance imaging (MRI), while surgery occurs in supine position, introducing spatial discrepancies due to breast deformation [6,7]. Current localization of non-palpable lesions includes wire-guided, carbon tattooing, clips, and seeds, which are invasive, uncomfortable, and lack optimal precision [4,8]. Supine MRI has also been proposed to better approximate the patient's surgical position [9,10].

Digital twins (DTs) are patient-specific virtual models derived from imaging that can be aligned with anatomy and visualized intra-operatively using extended reality (XR) [11–14]. They enable non-invasive, real-time guidance and have shown promise in other

surgical fields [12–16], though applications in breast surgery remain limited [17–19].

In this pilot study, we describe an XR-based DT navigation system for non-invasive tumor localization during BCS and assess its technical feasibility, usability, and integration into the surgical workflow.

2. Methods

2.1. Ethical approval

Four patients undergoing breast surgery were prospectively included after informed consent. The study was approved by the Ethics Committee of the Champalimaud Foundation in March 2022 with the identification *Breast 4.0 v 3.0* (no applicable approval number). All the patients were followed by the same surgeon who also performed the procedures described in this study.

* Corresponding author. Breast Cancer Research Program, Champalimaud Foundation, Av Brasília, 1400-038, Lisboa, Portugal.

** Corresponding author. Breast Cancer Research Program, Champalimaud Foundation, Av Brasília, 1400-038, Lisboa, Portugal.

E-mail addresses: tiago.marques@research.fchampalimaud.org (T. Marques), pedro.gouveia@fundacaochampalimaud.pt (P. Gouveia).

¹ co-senior authors (equal contribution).

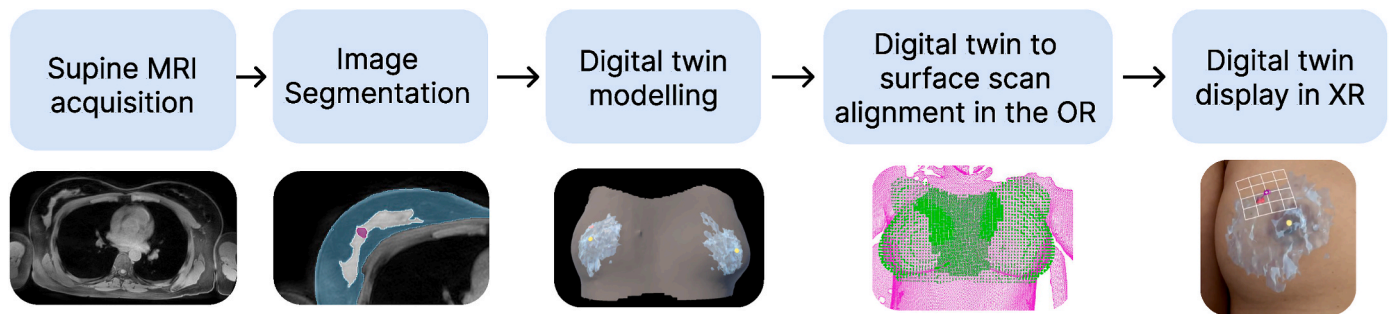


Fig. 1. From breast MRI to extended reality (XR) display in the operating room (OR). Breast tissues were segmented from supine MRI scans using 3D Slicer to generate a breast digital twin (DT). Afterwards, the DT was modelled using the Open3D library and custom-made software. In the OR, the DT was registered to real-time surface scans using two Azure Kinect sensors and displayed to the surgeon using an XR device (Magic Leap 2).

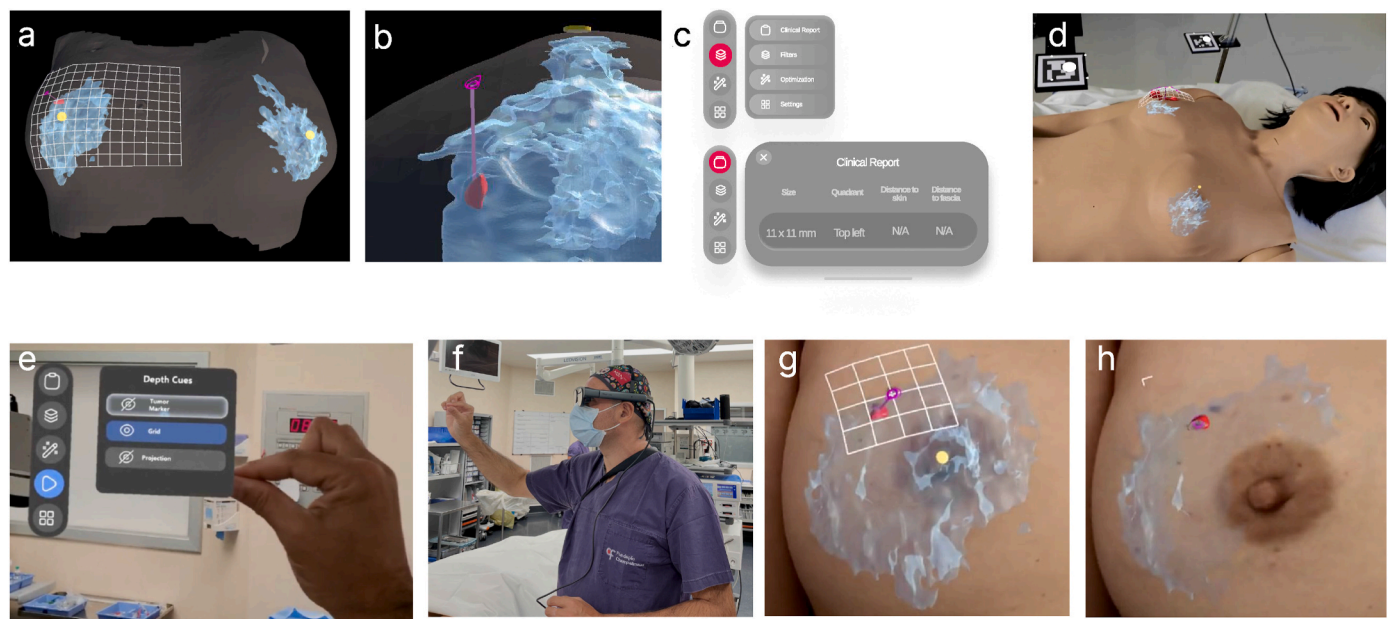


Fig. 2. XR interface development. a) DT with the final color scheme, including visual guides: fibroglandular tissue (in blue), tumor (in red), tumor projection (in pink), and a 1 cm spaced grid (in white). b) Side view of the tumor vertical projection on the breast: a pink line going from the center of the lesion to the skin surface, with the surface collision displayed as a pink circle. c) User interface menu early prototype design. d) DT aligned with the medical mannequin in co-design sessions. e) Final user interface with head-gaze and pinch gesture. f) Surgeon interacting with the user interface. g) Non-invasive tumor localization in the OR in a patient's breast. h) Control of DT visualization by moving closer resulting in digital contents being hidden. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2.2. Patient-specific digital twins for breast cancer surgery

Patient-specific DTs were generated from contrast-enhanced supine MRI scans, acquired using a 3T system and a 32-channel torso coil to minimize anatomical distortion. Parameters included: TE = 1.57 ms; TR = 5.3 ms; FOV = $450 \times 450 \text{ mm}^2$; 1.5 mm slice thickness; and a total scan time = 352 s. Skin, fibroglandular tissue, and tumor were segmented using 3D Slicer (Fig. 1). Post segmentation, the 3D model was processed using the Open3D library to extract the surface mesh and remove statistical outliers (Supplementary Fig. 1).

2.3. Optimized XR user interface

The XR interface and DT visualization scheme was co-designed with breast surgeons for the Magic Leap 2 (ML) XR headset and implemented in Unity 3D using MRTK3 (Fig. 2a–e). Head-gaze with pinch motion was chosen as the interaction scheme (Fig. 2f). The ML near clipping boundary ($\sim 37 \text{ cm}$) was intentionally used to control the digital content visualization, as the surgeon moves closer or further to the patient (see

Fig. 2g and h).

2.4. 3D sensing system for DT to patient-alignment

In the OR, a real-time surface scanning system composed of two Azure Kinect sensors was used to acquire surface scans of the patient (15 fps), keeping up with the patient's breathing motion [20]. The DT was automatically registered to the patient using an interactive closest point (ICP) algorithm and visualized through the ML, allowing the surgeon to assess tumor position relative to the breast surface (Supplementary Fig. 2).

3. Results

3.1. Cohort

The cohort ($n = 4$; median age 48 years) included heterogeneous tumor presentations, including a large lesion requiring mastectomy and a multifocal case (Fig. 3). Tumor volume (median = 0.25 cm^3 ; range =

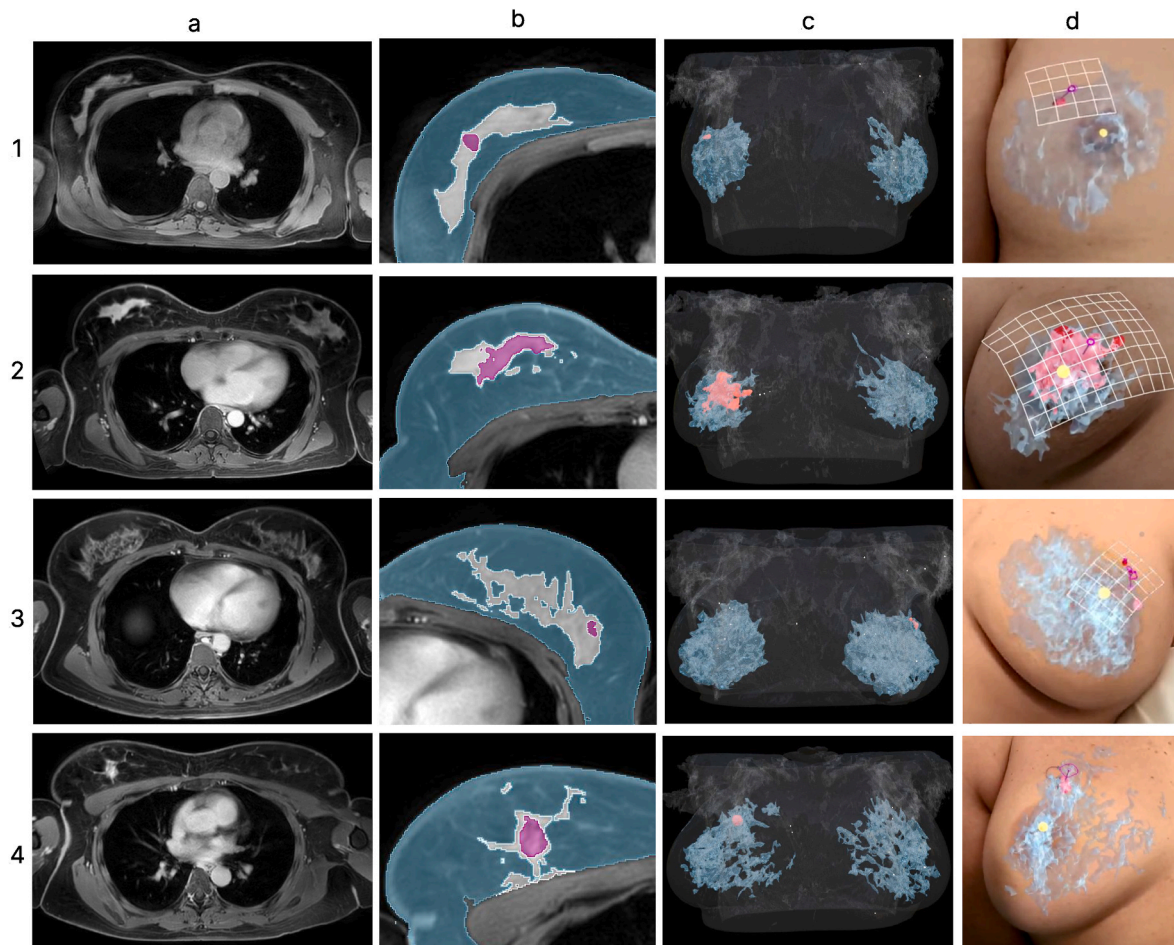


Fig. 3. DTs were created from supine breast MRI. a) Raw supine MRI images for the four patients in the study (one in each row). b) Segmented supine MRI images: fat (blue), fibroglandular tissue (white), and lesions (pink). c) DTs produced from the segmented MRIs: fat (white/grey), fibroglandular tissue (blue), and lesions (red). d) Images taken from live captures performed through the XR headset (ML) of the patients' DTs aligned with the torso in the OR for the four surgeries. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

[0.02, 12.74] cm³) and oriented bounding box (OBB) dimensions were derived from breast MRI segmentations (Supplementary Table 1).

3.2. Tumor marking in the operating room

System setup was completed in a median time of 218 s (range 72–254). Tumor marking required 49 s (range 32–72), without disrupting surgical workflow. Nipple deviation between the DT and patient ranged from 0 to 5 mm (Supplementary Table 2), indicating good surface alignment (Fig. 3d–Supplementary video). Tumor skin projection discrepancies relative to carbon tattoo markings varied substantially, ranging from minimal differences to >30 mm (Supplementary Table 2). Higher misalignment occurred when the tumor was located on the outer side of the breast, and the carbon tattoo marking was performed laterally rather than vertically. In these cases, the surgeon reported that intraoperative findings were consistent with the DT location, although no margin outcomes assessments were performed.

3.3. Questionnaires

The System Usability Score (1–100 scale) was 95 (range 92.5–10), rated as “Best Imaginable” in the adjective scale [21]. NASA-TLX scores revealed a very high performance (score = 100; range 100–100) with manageable mental and physical demands (Supplementary Table 3) [21–24]. In the satisfaction questionnaires, the surgeon rated the depth perception and interface intuitivity as excellent

(Supplementary Table 4).

4. Discussion and conclusion

This technical pilot study demonstrates that real-time alignment of patient-specific DTs is technically feasible and compatible with surgical workflows. By overlaying the tumor directly onto the patient using XR, this approach enables intuitive, non-invasive tumor localization, addressing limitations of conventional techniques, which are invasive and rely on 2D imaging interpretation [4,5,12,15].

Previous XR-based systems for breast surgery have been limited by 2D visualization or lack of real-time integration [17–19,25]. In other surgical fields, XR has shown clinical value, but most systems rely on manual or marker-based registration, leading to an accuracy decline over time due to intraoperative tissue deformation [26,27]. In BCS, this challenge is greater due to the lack of stable anatomical landmarks. Although markerless registration methods are under research, they currently underperform marker-based approaches, limiting their use in real-time clinical settings [28].

Although limited by the small cohort (n = 4) evaluated in this study, tumors varied considerably in volume, and included unifocal and multifocal cases. The system achieved nipple deviations ranging 0–5 mm, with minimal setup time (218 s) and tumor marking time (49 s). The surgeon reported a match between DT tumor position and the intraoperative findings. Usability testing showed high surgeon satisfaction and low perceived workload. These findings should be interpreted

as preliminary results, since this study focused mostly on technical feasibility and usability rather than ground-truth accuracy, as conventional localization methods lack standardization and have several limitations. Surgeon confirmation remained essential, and future studies should include larger, more homogeneous cohorts and evaluate end-points such as margin status.

A major unresolved challenge is breast tissue deformation. Although supine MRI improves anatomical correspondence compared to prone imaging, it does not fully replicate intraoperative conditions [29–31]. In our cohort, nipple and skin alignment was relatively high suggesting that these tissue deformations were limited. However, advanced biomechanical modeling and non-rigid registration methods capable of accounting for patient positioning, respiratory motion, and surgical manipulation deformations are likely to be necessary for clinical translation [32–34].

While manual image segmentation remains the gold standard for clinical applications, automated imaging processing pipelines may prove essential for scalability [35–37]. Beyond technical improvements, ethical considerations, social impact, legal guidance, and regulatory compliance (European Union Medical Device Regulation 2017/745) remain critical considerations for safe and responsible clinical implementation [12].

In conclusion, DT-based XR guidance for tumor localization is technically feasible and integrates well into surgical workflows. However, further technical development and clinical validation are required before routine clinical adoption.

CRedit authorship contribution statement

Rafaela Timóteo: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Alexandre Laborde:** Writing – review & editing, Software, Methodology. **Bruno Vaz:** Writing – review & editing, Software, Methodology. **Nuno Loução:** Writing – review & editing, Methodology. **Yasna Forghani:** Writing – review & editing, Methodology. **André Cardoso:** Writing – review & editing, Data curation. **Mariana Correia:** Writing – review & editing, Data curation. **João Santinha:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Daniel Simões Lopes:** Writing – review & editing, Supervision. **Tiago Marques:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Pedro Gouveia:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Code availability

The underlying code for the XR system and DTs alignment in the OR is not publicly available for proprietary reasons.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Tiago Marques, Pedro Gouveia, and Joao Santinha have patent METHODS AND SYSTEMS FOR PRECISION-GUIDED SURGERY pending to Champalimaud Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research is part of the Health from Portugal (HfP), funded by Agendas Mobilizadoras para a Inovação Empresarial - Plano de Recuperação e Resiliências (PRR) Português (no applicable grant number). The funder played no role in study design, data collection, analysis and

interpretation of data, or the writing of this manuscript. The authors would like to thank the Breast Unit at Champalimaud Clinical Center for their cooperation in this study.

Appendix A. Supplementary data

Supplementary data to this article can be found at <https://doi.org/10.1016/j.breast.2026.104807>.

Data availability

The breast supine MRIs, tissue segmentations, and DTs for the four patients are available in this study repository at <https://tinyurl.com/3bedrhpa>.

References

- [1] The International Agency for Research on Cancer (IARC). (n.d.). Global cancer observatory. <https://gco.iarc.who.int/en>. Last accessed 2024/February/8..
- [2] Cardoso F, Kyriakides S, Ohno S, Penault-Llorca F, Poortmans P, Rubio IT, Senkus E. Early breast cancer: ESMO clinical practice guidelines for diagnosis, treatment and follow-up. *Ann Oncol* 2019;30(8):1194–220.
- [3] Yu CX. Radiotherapy of early-stage breast cancer. *Precis Radiat Oncol* 2023;7(1): 67–79.
- [4] Corsi F, Sorrentino L, Bossi D, Sartani A, Foschi D. Preoperative localization and surgical margins in conservative breast surgery. *Int J Surg Oncol* 2013;2013: 793819.
- [5] Ryan JF, Ghosh S, Rajae AN, Olson D, Lesniak DM, Peiris LJ, Alberta Breast Surgical Oncology Research Group. Re-excision rates after breast-conserving surgery for invasive breast cancer: an albertan perspective. *Can J Surg J Can Chir* 2024;67(6):E363–9. <https://doi.org/10.1503/cjs.004723>.
- [6] Richey WL, Heiselman JS, Ringel MJ, Meszoely IM, Miga MI. Computational imaging to compensate for soft-tissue deformations in image-guided breast conserving surgery. *IEEE (Inst Electr Electron Eng) Trans Biomed Eng Dec*. 2022;69 (12):3760–71. <https://doi.org/10.1109/TBME.2022.3177044>.
- [7] Bessa S, Gouveia PF, Carvalho PH, Rodrigues C, Silva NL, Cardoso F, Cardoso MJ. 3D digital breast cancer models with multimodal fusion algorithms. *Breast* 2020; 49:281–90.
- [8] Garzotto F, Comoretto RI, Michieletto S, Franzoso G, Mele ML, Gregori D, Saibene T. Preoperative non-palpable breast lesion localization, innovative techniques and clinical outcomes in surgical practice: a systematic review and meta-analysis. *Breast* 2021;58:93–105.
- [9] Barth Jr RJ, Krishnaswamy V, Paulsen KD, Rooney TB, Wells WA, Angeles CV, Zuurbier RA, Rosenkranz K, Poplack S, Tosteson TD. A randomized prospective trial of supine MRI-guided versus wire-localized lumpectomy for breast cancer. *Ann Surg Oncol* 2019;26(10):3099–108. <https://doi.org/10.1245/s10434-019-07531-4>.
- [10] Fausto A, Fanizzi A, Volterrani L, Mazzei FG, Calabrese C, Casella D, Marcasciano M, Massafra R, La Forgia D, Mazzei MA. Feasibility, image quality and clinical evaluation of contrast-enhanced breast MRI performed in a supine position compared to the standard prone position. *Cancers* 2020;12(9):2364. <https://doi.org/10.3390/cancers12092364>.
- [11] Zhang K, Zhou HY, Baptista-Hon DT, Gao Y, Liu X, Oermann E, Wu J. Concepts and applications of digital twins in healthcare and medicine. *Patterns* 2024;5(8).
- [12] Mekki YM, Luijten G, Hagert E, et al. Digital twins for the era of personalized surgery. *npj Digit Med* 2025;8:283. <https://doi.org/10.1038/s41746-025-01575-5>.
- [13] Shi Y, Deng X, Tong Y, Li R, Zhang Y, Ren L, Si W. Synergistic digital twin and holographic augmented-reality-guided percutaneous puncture of respiratory liver tumor. *IEEE Trans Hum-Mach Syst* 2022;52(6):1364–74.
- [14] Kolekar A, Shalgar S, Malawade I. Beyond reality: a study of integrating digital twins. In: *Journal of physics: conference series*, 2601. IOP Publishing; 2023, September, 012030. 1.
- [15] Katsoulakis E, Wang Q, Wu H, et al. Digital twins for health: a scoping review. *npj Digit Med* 2024;7:77. <https://doi.org/10.1038/s41746-024-01073-0>.
- [16] Asciak L, Kyeremeh J, Luo X, et al. Digital twin assisted surgery, concept, opportunities, and challenges. *npj Digit Med* 2025;8:32. <https://doi.org/10.1038/s41746-024-01413-0>.
- [17] Gouveia PF, Costa J, Morgado P, Kates R, Pinto D, Mavioso C, Cardoso MJ. Breast cancer surgery with augmented reality. *Breast* 2021;56:14–7.
- [18] Sharifian R, Madad-Zadeh S, Bourdel N, Giro A, Marraoui W, Pomel C, Bartoli A. Guiding breast conservative surgery by augmented reality from preoperative MRI: Initial system design and retrospective trials. In: *Deep breast workshop on AI and imaging for diagnostic and treatment challenges in breast care*. Cham: Springer Nature Switzerland; 2024, October. p. 212–20.
- [19] Lee M, Woo J, Peak SH, Kim HG, Lim WS, Chung J, Lee JW. An exploratory clinical trial of preoperative non-invasive localization before breast-conserving surgery using augmented reality technology. *Breast Cancer Res Treat* 2024;206(1):31–44.
- [20] Timóteo R, Laborde A, Forghani Y, Santinha J, Lopes DS, Gouveia P, Marques T. Automatic and high-resolution surface scan system for breast cancer treatment: clinical accuracy and usability assessment. *Eur J Surg Oncol* 2026;52:111507.

- [21] Bangor A, Kortum P, Miller J. Determining what individual SUS scores mean: adding an adjective rating scale. *J Usability Stud* 2009;4(3):114–23.
- [22] Hart SG, Staveland LE. Development of NASA-TLX (task load index): results of empirical and theoretical research. In: *Advances in psychology*, 52; 1988. p. 139–83. North-Holland.
- [23] Byers JC, Bittner AC, Hill SG. Traditional and raw task load index (TLX) correlations: are paired comparisons necessary. *Adv Ind Ergon Saf* 1989;1:481–5.
- [24] Brooke J. SUS-A quick and dirty usability scale. *Usability Eval Ind* 1996;189(194): 4–7.
- [25] Ock J, Moon S, Kim M, Ko BS, Kim N. Evaluation of the accuracy of an augmented reality-based tumor-targeting guide for breast-conserving surgery. *Comput Methods Progr Biomed* 2024;245:108002.
- [26] Lan T, Liu S, Dai Y, Luo J, Han J, Jin Y. Real-time image fusion and Apple Vision Pro in laparoscopic microwave ablation of hepatic hemangiomas. *npj Precis Oncol* 2025;9(1):79.
- [27] Javaheri H, Ghamarnejad O, Lukowicz P, Stavrou GA, Karolus J. From concept to clinic: multidisciplinary design, development, and clinical validation of augmented reality-assisted open pancreatic surgery. In: *Proceedings of the 2025 CHI conference on human factors in computing systems*; 2025, April. p. 1–24.
- [28] Tu M, Jung H, Moghadam A, Raythatha J, Hsu J, Kim J. Exploring the performance of geometry-based markerless registration in a simulated surgical environment: a comparative study of registration algorithms in medical augmented reality. In: *2023 45th annual international conference of the IEEE engineering in medicine & biology society (EMBC)*, Sydney, Australia; 2023. p. 1–4. <https://doi.org/10.1109/EMBC40787.2023.10341197>.
- [29] Navab N, Martin-Gomez A, Seibold M, Sommersperger M, Song T, Winkler A, Eck U. Medical augmented reality: definition, principle components, domain modeling, and design-development-validation process. *J Imaging* 2022;9(1):4.
- [30] Gouveia PF, Luna R, Fontes F, Pinto D, Mavioso C, Anacleto J, Cardoso MJ. Augmented reality in breast surgery education. *Breast Care* 2023;18(3):182–6.
- [31] Riedel P, Riesner M, Wendt K, Abmann U. Data-driven digital twins in surgery utilizing augmented reality and machine learning. In: *2022 IEEE international conference on communications workshops (ICC workshops)*. IEEE; 2022, May. p. 580–5.
- [32] Carolina L, Timóteo R, Laborde A, Vaz B, Gouveia P, Marques T. Volumetric correction for aligning breast digital twins with surface scan. In: *Proceedings of the 14th conference on new technologies for computer/robot assisted surgery*; 2025.
- [33] Teixeira AM, Martins P. A review of bioengineering techniques applied to breast tissue: mechanical properties, tissue engineering and finite element analysis. *Front Bioeng Biotechnol* 2023;11:1161815. <https://doi.org/10.3389/fbioe.2023.1161815>.
- [34] Eiben B, Vavourakis V, Hipwell JH, Kabus S, Buelow T, Lorenz C, Mertzaniadou T, Reis S, Williams NR, Keshtgar M, Hawkes DJ. Symmetric biomechanically guided prone-to-supine breast image registration. *Ann Biomed Eng* 2016;44(1):154–73. <https://doi.org/10.1007/s10439-015-1496-z>.
- [35] Forghani Y, Timóteo R, Marques T, Loução N, Cardoso MJ, Cardoso F, Santinha J. Comparative analysis of nnU-Net and Auto3Dseg for fat and fibroglandular tissue segmentation in MRI. *J Med Imaging* 2025;12(2). 024005-024005.
- [36] Lew CO, Harouni M, Kirksey ER, et al. A publicly available deep learning model and dataset for segmentation of breast, fibroglandular tissue, and vessels in breast MRI. *Sci Rep* 2024;14:5383. <https://doi.org/10.1038/s41598-024-54048-2>.
- [37] Narimani S, Roth Hoff S, Dæhli Kurz K, et al. Comparative analysis of deep learning architectures for breast region segmentation with a novel breast boundary proposal. *Sci Rep* 2025;15:8806. <https://doi.org/10.1038/s41598-025-92863-3>.